

Ch5_Data Sources

重点: Internal /External data sources (dis)advantages

What is Data sources?

A data source is the origin or location from which data is extracted for analysis, processing, or display

Internal Data Sources

Internal data sources are data that the **organization controls**.

- Stored and managed within the organization.
- Data:
 - Proprietary 专有 – 拥有独立产权的数据
 - Bought – 第三方市场购买（整合进组织内部系统/完全掌控数据
 - Collected

Sales/marketing data	包括客户 <u>购买记录</u> 、广告点击率、市场活动反馈等， 用于制定营销策略。
Finance data	包括 <u>收入、支出、利润报表</u> 等，用于财务分析与对外报告。
Production data – supplier	包括供应链信息、ERP系统中的产品价格与库存数据，用于生产计划与成本控制。
Human resources data	包括员工信息、绩效记录、招聘数据等，用于人力资源管理与组织发展。

Different types of Internal source

1. **Sales/Marketing** Data and Systems

Plan, promotion based on costumer data.

- Customer Data – 包括人口统计信息（demographics）、偏好（preferences）、行为（behaviors）

- Sales Data – 销售管道 (pipeline)、周期 (cycle)、交易量 (volume)、收入 (revenue)、预测 (forecasts)
- Marketing Data – 活动 (campaigns)、销售渠道 (channels)、投资回报率 (ROI)、指标 (metrics)

Pipeline (销售管道)：潜在客户从接触到成交的流程阶段。



这是一个**流程模型**，但在 CRM 系统中，每个阶段都会被**数据化**

Cycle (销售周期)：完成一次销售所需的时间。

ROI (投资回报率)：衡量市场活动的投入产出比

Metrics (指标)：如点击率、转化率、客户获取成本等。

- System:

- general name
- CRM systems (e.g., Salesforce, HubSpot)
 - Marketing **automation** systems (e.g., Mailchimp, Marketo)
 - Web analytics systems (e.g., Google Analytics, Adobe Analytics)
- 分析用户针对于website的行为 (如click, scroll through..)

客户关系管理系统 (CRM)	管理客户信息、销售流程、客户互动记录。支持销售团队协作与客户生命周期管理。
市场自动化系统	自动执行市场活动，如邮件营销、潜在客户培育。强调自动化的重要性。
网站分析系统	跟踪用户行为、流量来源、页面表现。用于优化网站内容与用户体验。

2. Finance Data and system

Financial report – report to stakeholder → generated from the system

- Related to money

Revenues	收入	企业通过销售产品或服务获得的 总金额 。
Expenses	支出	企业运营过程中发生的 成本 ，如工资、租金、采购等。
Assets	资产	企业拥有的 资源 ，如现金、设备、库存等。
Liabilities	负债	企业需偿还的 债务 ，如贷款、应付账款。
Cash Flows	现金流	企业现金的流入与流出情况。资金调度情况

Budgets	预算	对未来收入与支出的计划安排。
Forecasts	预测	对未来财务表现的预估，如销售预测、利润预测。
Financial Ratios	财务比率	如利润率、资产负债率等。

ERP system	集成企业核心流程，如财务、采购、库存管理。强调“集中化平台”与“自动化”。
Business Intelligence system	可视化分析财务数据，生成图表与仪表板。用于决策支持与趋势洞察。
Financial planning and analysis system	用于预算编制、财务预测与模拟分析。支持战略财务管理。

- ERP systems (e.g., SAP, Oracle)
- Business intelligence systems (e.g., Tableau, Power BI)
- Financial planning and analysis systems (e.g., Adaptive Insights, Anaplan)

ERP, or **Enterprise Resource Planning**, is a type of software that integrates and manages a company's core business processes into a single, unified system. It streamlines functions like finance, human resources, inventory, and supply chain management, helping organizations improve efficiency, reduce costs, and make better data-driven decisions. [🔗](#)

3. Production Data and systems

- Manufacturing or creation of goods and service data.
- Supplier

Production Volume 产量	指单位时间内生产的产品数量
Costs	包括原材料、人力、设备等成本，
Quality	产品符合标准的程度，用于质量管理与客户满意度提升。
Efficiency	资源利用率，如单位时间产出，用于流程优化。
Capacity 产能	最大可生产数量

Systems:

- MES (Manufacturing Execution System) systems (e.g., Siemens, Rockwell Automation)
 - SCADA (Supervisory Control and Data Acquisition) systems (e.g., Wonderware, Inductive Automation)
 - ERP systems for production management
-
- MES – 实时监控生产过程，连接车间与管理层，实现生产调度与质量追踪。
 - SCADA – 采集传感器数据，控制生产设备，适用于工业自动化场景。
 - ERP – 企业资源计划系统 -> 管理生产计划、库存、采购等，与其他部门系统集成

4. Human Resource Data and Systems

Recruitment 招聘	包括职位发布、简历筛选、面试安排等，用于人才获取。
Retention 留任	员工保留率与流失率分析，用于组织稳定性评估。
Performance (KPI)	用于评估员工工作成果，如销售额、项目完成率。
Compensation	包括工资、奖金、福利等，用于薪酬管理与激励机制。
Training	员工技能提升与发展计划，用于人才培养与合规要求。

Also, HR retain employee by benchmarking with their parnter (external) – whether or not increase salary ->了解市场内同类型职务工资

HRIS (人力资源信息系统)	管理员工基本信息、入职流程、考勤记录等。
HCM (人力资本管理系统)	更全面的人力资源战略管理，包括绩效、继任计划、组织架构。
员工参与系统 (Engagement)	收集员工反馈、满意度调查，用于文化建设与员工体验优化。

- HRIS (Human Resource Information System) systems (e.g., BambooHR, Workday)
- HCM (Human Capital Management) systems (e.g., Oracle HCM Cloud, SAP SuccessFactors)
- Employee engagement systems (e.g., Culture Amp, Glint)

Advantage

Increased reliability

Direct control over data collection and management

Higher accuracy and credibility

Direct control --> High accuracy --> reliability

Better security

Restricted access to authorized personnel

Enhanced data protection measures

Restricted access + protection --> Security

Easily accessible

Immediate access for quick decision-making

Integration with existing systems

Easy accessible --> Immediate access --> quick decision-making

Disadvantage

Limited scope

Focused only on company's transactions and practices

May not provide complete market or industry picture

Only company + x complete market --> Limited scope
--> Bias

Biased perspective

Specific to company operations

May not reflect broader trends

collection + maintenance + ensure consistencies --> costly + time
-consuming

Costly and time-consuming

Resources required for collection and maintenance

Potential for data silos and inconsistencies

Data silos are isolated collections of data that are trapped within specific departments or systems, making them inaccessible to the rest of the organization

- They often form unintentionally as different departments (like marketing, finance, or HR) use their own specific tools and applications to meet their unique needs.

=> Inefficiency, Inconsistent, Poor decision-making, reduced productivity...

External data source

Data originating from outside the company

Provides broader context and market insights
 Supplements internal data for comprehensive analysis

商业平台数据	Job portals and career websites	招聘网站、职业平台，如 LinkedIn。
学术	Patents and university data	专利数据库、高校研究成果
政府与统计数据	Public statistics and government sources	政府发布的统计数据，如人口普查、经济指标。
社会行为数据	Social media and consumer transactions	社交媒体行为、消费者交易记录，如 Twitter、电商平台数据。

情境示例	数据来源	是否属于 external
企业从电商平台（如淘宝、京东）获取行业交易数据	第三方平台	✅ 是
企业购买消费者行为报告（如尼尔森、QuestMobile）	数据供应商	✅ 是
企业通过 API 获取社交平台上的购买行为	外部系统	✅ 是

如果交易发生在企业自有平台（如自营电商网站、POS系统），并由企业存储和管理，那就是 internal。

Advantages

Increased scope
 Broader range of information from various sources
 Insights into market trends and competitor activities
 various sources -> larger scope
 [market trend + competitor activities]

Reduced costs
 Often less expensive than generating internal data
 Access to large-scale datasets

independent third-party data collection -> bias ↓ -> accuracy ↑

Improved accuracy
 Independent third-party data collection
 Reduced internal bias

independent third-party data collection（独立第三方数据收集）– 由不属于数据使用方的独立机构或平台进行的收集、整理、加工数据活动

相对于数据使用方，不存在 internal bias

Disadvantage

Limited relevance

May not directly apply to specific business needs

Challenges in interpretation and application

Limited quality

Varying standards and practices in data collection

Potential for inaccuracies or inconsistencies

Limited accessibility

Legal, ethical, or technical restrictions

Potential costs for premium data sources

1. Limited Relevance (相关性有限)

May not directly apply to specific business needs	可能与具体业务需求不直接相关
Challenges in interpretation and application	在理解与应用上存在困难

- 某些数据虽然真实，但可能不适用于当前业务场景，例如行业数据太宽泛，无法支持具体决策。
- 数据结构或术语不一致，导致解释困难，如外部数据使用不同的分类标准。

2. Limited Quality (质量有限)

Varying standards and practices in data collection	数据收集标准与方法不一致
Potential for inaccuracies or inconsistencies	可能存在不准确或不一致性

- 不同机构使用不同的采集方式（如问卷 vs 爬虫），导致数据标准不统一。
- 数据可能存在缺失值、错误值、重复记录，影响分析结果。

3. Limited Accessibility (可访问性有限)

Legal, ethical, or technical restrictions	法律、伦理或技术限制
Potential costs for premium data sources	高质量数据可能需要付费

- 某些数据受限于隐私法规（如GDPR）或商业协议，无法自由使用。
- 高质量数据（如行业报告、实时交易数据）通常需要购买或订阅，增加成本。

非重点内容

Data Collection Methods

Qualitative Methods -> In-depth

- Interviews: In-depth conversations with participants 1-1
- Focus groups: Moderated group discussions Group
- Observations: Systematic recording of behaviors or phenomena

Quantitative Methods -> Data

- Surveys: Structured questionnaires for large samples
- Experiments: Controlled studies to test hypotheses
- Observations with numerical data: Quantified behavioral data

Mixed Methods Research

- Convergent parallel design
 - Collect and analyze quantitative and qualitative data separately
 - Compare results to identify similarities and differences
- Embedded design
 - Collect both types of data within a larger quantitative or qualitative design
 - Use qualitative data to provide in-depth understanding of quantitative findings
- Explanatory sequential design
 - Collect quantitative data first, followed by qualitative data
 - Use qualitative data to explain quantitative findings
- Exploratory sequential design
 - Collect qualitative data first, followed by quantitative data
 - Use quantitative data to test or confirm qualitative findings



Convergent Parallel	同时收集	并行分析后比较结果
Embedded	主体中嵌入另一种数据	辅助解释主数据
Explanatory Sequential	先定量后定性	定性解释定量结果
Exploratory Sequential	先定性后定量	定量验证定性发现

1. Convergent Parallel Design -> 并行收敛设计
2. Embedded Design -> 嵌入式设计

- 当定量研究为主，定性数据作为补充说明。
- 同步/嵌入式

定量为主	嵌入定性数据	问卷调查 + 开放式访谈	用访谈解释问卷结果中的异常
定性为主	嵌入定量数据	深度访谈 + 小规模量表	用量表验证访谈中提到的趋势

3. Explanatory Sequential Design -> 解释性顺序设计

- 当你发现定量结果有“为什么”的问题时，用定性方法补充解释。

4. Exploratory Sequential Design -> 探索性顺序设计